

AI Sales Operations Agent

Current business capabilities, technical architecture, validated workflows, controls and production-readiness assessment

BUSINESS SCOPE	TECHNICAL FOUNDATION	VALIDATED OUTCOME
Inquiry capture through quotation, follow-up, negotiation, PO validation, sales order and department handoff.	FastAPI, LangGraph, Gemini, local Qdrant, local embeddings, SQLite, agent-scoped retrieval and checkpoints.	A live MSB-076 scenario reached handed_off with production, inventory, purchase, dispatch and finance notified.

Executive position

The system is a functional local sales-operations orchestrator for an industrial B2B workflow. It combines LLM reasoning for language tasks with deterministic calculations and exact structured-document retrieval for inventory, capacity, delivery, pricing, margin, tax, transport and discount decisions. It has been validated from inquiry through final sales-order handoff in the local test environment.

Report basis

- Repository implementation and uploaded demo-steel-company document pack.
- Live API validation performed on 24 July 2026.
- Local test data includes synthetic product-cost and normalized discount records.

Current maturity

Functional prototype	Controlled local pilot	Production hardening required
Completed	Suitable with supervised users	Before unsupervised customer use

1. Business capabilities

The agent covers the main commercial lifecycle of an industrial B2B sales inquiry. It is designed to assist sales operations, not replace authorized human decisions.

Capability	What the agent does	Business value
Inquiry capture	Normalizes email or message text, extracts customer, company, product, quantity, specifications, delivery and payment expectations, and requests missing information.	Reduces manual data entry and incomplete leads.
Requirement matching	Matches a request to catalog evidence, classifies exact, near or custom, identifies specification gaps and flags technical review.	Prevents unsupported products from being quoted as standard items.
Customer qualification	Looks up customer history, payment behavior, credit position, order history and lead quality, then assigns score, temperature and priority.	Helps sales teams prioritize and identify credit risk.
Feasibility	Checks stock, available production capacity, delivery zone, transit time, lead time and deadline risk.	Avoids commitments that operations cannot fulfill.
Pricing	Calculates RM cost, overhead, transport, floor price, list price, discount, margin, GST and total invoice value.	Creates auditable prices and blocks incomplete pricing data.
Quotation	Builds a structured quotation draft, renders HTML, assigns payment terms, delivery timeline, validity and approval reasons, and saves the record.	Standardizes quote quality and commercial terms.
Follow-up	Generates scheduled reminder messages with escalating tone and logs follow-up attempts.	Improves response discipline and pipeline coverage.
Negotiation	Classifies replies, detects objections and counteroffers, compares offers with floor price, revises quotations or rejects unsafe offers.	Protects margin while accelerating negotiation response.
PO validation	Extracts PO fields and compares product, quantity, rate, GST, total, payment and delivery terms with the latest quotation.	Reduces order-entry errors and commercial leakage.
Sales order and handoff	Marks the lead won, creates a sales order, prepares department packages and notifies production, inventory, purchase, dispatch and finance.	Provides a controlled transition from sales to execution.

2. End-to-end operating model

A stable thread ID keeps one lead's state across separate customer and internal events. The graph is re-entered with a trigger rather than rerunning the entire sales lifecycle from the beginning.

Inquiry	Requirement	Qualification	Feasibility
Pricing	Quotation	Follow-up	Reply / negotiation
PO validation	Sales order	Handoff	

Event triggers

Trigger	Pipeline entered	Typical final state
inquiry	Capture through quotation	quotation_sent or a controlled gate
followup	Reminder composition and dispatch	followup_sent
customer_reply	Reply analysis, objection or counteroffer	message dispatched, revised quote or approval
po_received	PO extraction, validation and order handoff	handed_off, correction request or PO approval
approved	Resume the exact paused stage	Next stage or final outcome

Human control points

Qualification Credit-risk exceptions and uncertain customer exposure.	Feasibility Capacity, deadline, custom-product or delivery commitment exceptions.
Pricing Valid calculated prices that exceed approval thresholds, discounts or margin rules.	PO Internal review of minor or policy-sensitive PO mismatches.

Missing fundamental pricing inputs are now classified as blocked:pricing_data, not as an approvable pricing exception. This prevents human approval from bypassing a missing cost, tax, margin, discount or transport record.

3. Technical architecture

The solution uses a compiled LangGraph as the orchestration layer, FastAPI as the application boundary, Gemini for selected language tasks, Qdrant for document retrieval and SQLite for local operational records.

API layer	FastAPI endpoints for uploads, inquiries, events, approvals and diagnostics
Orchestration	LangGraph state machine with trigger routing and in-memory checkpoints
Agent layer	Specialized inquiry, catalog, qualification, inventory, feasibility, pricing, quotation, follow-up, negotiation, PO and handoff modules
Reasoning	Gemini structured output for extraction, classification, gap analysis, narrative and PO parsing
Knowledge	Local sentence-transformer embeddings and local persisted Qdrant collection
Structured data	Exact Qdrant payload-filter retrieval by business, agent, filename, type, active status and latest version
Persistence	SQLite for leads, audit logs, quotations, POs, sales orders and handoff records

Dual retrieval model

The architecture now separates semantic evidence retrieval from deterministic structured lookup. This corrects the earlier failure mode where top-K similarity search could retrieve a document but miss the exact required CSV row.

Semantic RAG	Exact structured retrieval
Used for product descriptions, technical evidence, customer notes, terms, templates, objection guidance and narrative grounding.	Used for product code, inventory, capacity, delivery city, price, product cost, margin, GST, transport zone and discount band.
Embedding query, allowed document policy, similarity score and top-K result.	Payload filters, no embedding score, complete document chunks and deterministic parsing.

4. Data, isolation and API surface

Documents are uploaded with business ownership, document type, agent access, version and active status. Agent policies restrict which document types each sub-agent may retrieve.

Data domain	Documents currently available	Decision key
Catalog	Product catalog and technical descriptions	Product code and specifications
Customer	CRM, payment behavior, order and quotation history	Company/customer identity
Operations	Inventory, capacity, delivery zones	Product code and normalized city
Commercial	Price, product cost, margin, GST, transport, discount and payment terms	Product, zone, customer type and order value
Documents	Quotation templates, terms and conditions	Agent-scoped semantic evidence

Current API surface

Endpoint	Purpose
GET /healthz	Reports Qdrant, RAG, graph, Gemini and database readiness.
POST /documents/upload	Saves, parses, chunks, embeds and indexes an authorized business document.
POST /rag/retrieve	Tests semantic agent-scoped retrieval.
POST /rag/document	Tests exact structured document retrieval without embeddings.
POST /inquiries/process	Creates a stable thread ID and runs the inquiry pipeline.
POST /pipeline/events	Processes follow-up, customer reply or PO events on an existing thread.
POST /pipeline/approve	Approves only the exact pending business stage and blocks unsafe pricing bypass.

State and auditability

- A thread ID correlates all events for one lead while the server remains running.
- The graph records completed stages, approval reasons, pipeline status and outbound content.
- RAG audit records capture agent, query, chunk IDs, document IDs and similarity scores.
- Pricing records retain calculation logic and missing-input validation details.
- Operational actions are written to the SQL audit log.

5. Validated scenario and controls

A live end-to-end scenario was executed against the local API after the exact structured retrieval upgrade.

Test input	Observed result
Product	MSB-076 - Steel Billet Model 076, IS2062 E250, size 100mm
Quantity and destination	20 MT to Delhi
Requirement	Exact match
Feasibility	From stock; delivery zone North; location found
Pricing	Pricing possible; final ex-GST price INR 104,870.44/MT; invoice INR 2,349,097.80
Quotation	Quotation generated and saved
PO	Minor mismatch on special conditions; PO approval requested
Final result	handed_off; order won; sales order and handoff IDs created
Departments	Production, inventory, purchase, dispatch and finance

Safety controls

Control	Current behavior
Missing inquiry data	Customer follow-up request and pipeline pause.
Unsupported/custom product	Technical or human review before pricing.
Insufficient fulfillment	Feasibility approval or inability-to-fulfill result.
Missing pricing inputs	blocked:pricing_data; no quotation and no approval bypass.
Large order/discount/margin	Pricing approval only after a valid price exists.
Critical PO mismatch	Customer correction request; order is not won.
Minor PO mismatch	Controlled PO approval before sales-order creation.

6. Current limitations and production roadmap

The system is suitable for a supervised local pilot. The following items should be completed before it sends binding commercial commitments without human oversight.

Priority	Limitation	Recommended action
High	MemorySaver checkpoints disappear when Uvicorn restarts.	Use PostgreSQL-backed LangGraph checkpoints and expose thread history.
High	Generated product-cost and normalized-discount data are synthetic test records.	Replace with approved BOM-based product costs and signed commercial policies.
High	Message dispatch is mocked.	Integrate authorized email/WhatsApp providers with delivery receipts and retry controls.
High	Local embedded Qdrant permits only one process to hold the storage lock.	Run Qdrant Server for concurrent workers and production deployment.
Medium	Document version selection uses active status and latest version naming.	Add an explicit uploaded_at field, effective dates and automatic deactivation of replaced versions.
Medium	Requirement matching remains LLM-assisted when an explicit deterministic catalog key is unavailable.	Add direct product-code extraction and normalized specification comparison before semantic fallback.
Medium	Natural-language dates and encoding may produce null deadlines or mojibake.	Normalize dates to ISO, enforce UTF-8 end to end and add locale-aware tests.
Medium	Approval is an API action without authenticated user identity or role enforcement.	Add authentication, RBAC, approver identity, comments, timestamps and immutable audit trails.
Medium	No automated regression suite currently proves every branch.	Add unit, graph-route, API, document-schema and end-to-end tests for all approval and mismatch cases.

Recommended roadmap

Phase 1 - Controlled pilot	Phase 2 - Operational integration	Phase 3 - Production scale
Use verified master data, supervised approvals, persistent checkpoints and automated regression tests.	Connect CRM/ERP, inventory, email/WhatsApp, identity and approval roles.	Deploy Qdrant Server, PostgreSQL, workers, monitoring, security controls and disaster recovery.

Bottom line

Today, the agent can coordinate a complete supervised sales workflow from incoming requirement to operational handoff, with auditable calculations and controlled approvals. Its strongest production feature is the separation of language reasoning from deterministic business data. Its next milestone is replacing local/test infrastructure and synthetic master data with persistent, authenticated and enterprise-integrated services.